

How to Build Baby Hoover V3

The design that meets the 38 requirements from the V2 analysis: a race model that knows where the race is, lines checked before they air, a camera that keeps the humans at the centre and presses keys safely, and instruments that can fail.

DOCUMENT	Hoover_T11_V3_Build_Paper_V1_2_16SEP26
BUILDS	T11_F125_Baby_Hoover_V3 (same tool lineage as V1 and V2)
ANSWERS	Hoover_T11_V2_Consolidated_Analysis_V1_16SEP26: requirements R-01 to R-38, defects D-20 to D-68
BUILDER	Claude Code: Pass 0 (harness), then two build passes in gated internal stages; then validation
OPERATOR	Mike, by card: one run per pass on the real captures, four in all
STATUS	V1.2: four passes instead of seven; detectors A29–A31 added. Pass 0 built and verified; Mike’s run underway.

1 Summary

V2 failed in one pattern. It turned each packet straight into a sentence or a cut, trusted the packet, and never checked afterwards. V3 puts a **race model** between the wire and everything that speaks or points. The model holds one clock, the state of the race, the state of every car, and facts derived from sustained evidence rather than single events. The Booth and Gallery read that model. Nothing reaches air without being re-checked against it at the moment it plays.

The build runs in four passes. **Pass 0 builds the test harness first**, and must prove it can fail: V2 run through it has to fail in exactly the places the analysis found. **Pass 1 builds the race model**: clock, race state and truth. **Pass 2 builds the broadcast**: camera, voice, audio kit and capture hygiene. **Pass 3 validates** on a live game and a new league race. Mike runs each pass once on the real captures.

V2’s protected capabilities carry over intact. V3 rebuilds the middle; it is not a restart.

What V3 is and is not

IN V3	NOT IN V3
Race and car state models, one clock, truthful lifecycle calls	Live text-to-speech on air
Derived overtakes, claims re-validated at air time	Language models: Toddler Hoover, after Baby Hoover (seam built now)
Stakes-aware Gallery, safe actuation, Focus Contract logged	Race Engineer, multi-client telemetry, mixed privacy, pre-race data
Lines in a data file; filler from real race material	Tuning by taste: the professional-broadcast ruler stays the bar
Timed-track kit for manual audio; wire-based test harness	Weather narrative beyond verified changes; OBS control changes

2 Design stance and decisions

Six rules the design follows

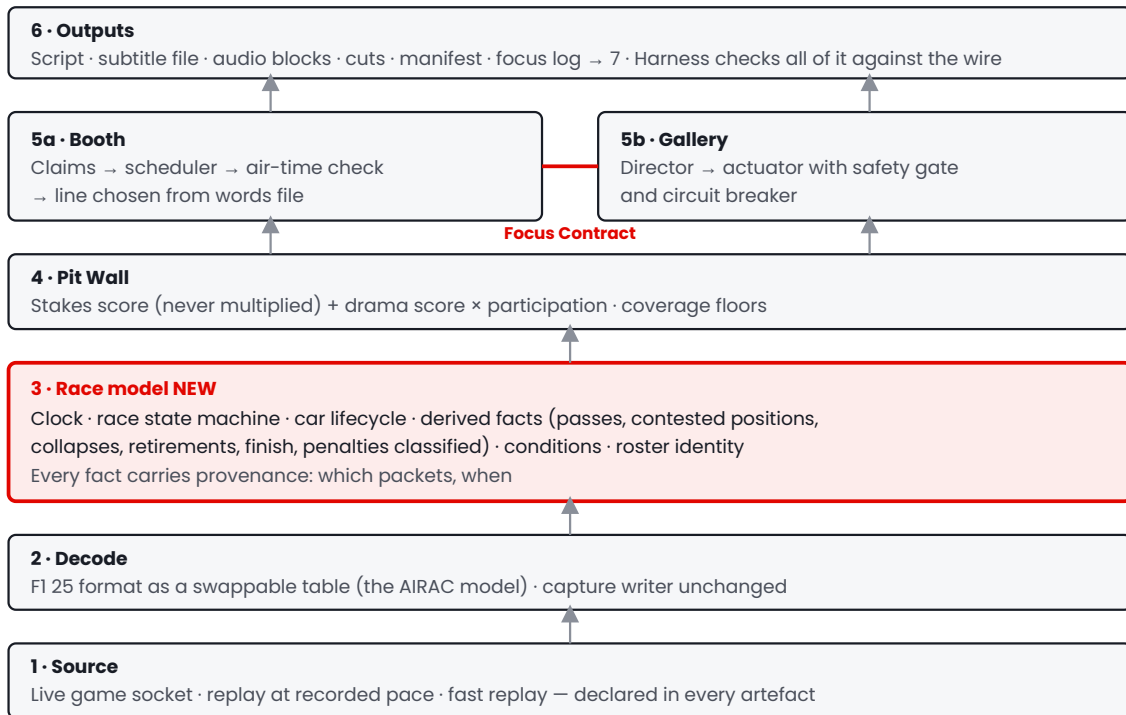
RULE	WHAT IT MEANS IN V3
State before speech	No component reacts to a packet directly. Packets update the race model; Booth and Gallery read the model.
Evidence before events	Game events are hints. Facts come from sustained evidence in lap data, session data and the final classification. This is the L5 finding (overtake events are unreliable) applied everywhere.
Checked at air time	A line is a claim with a validity test. The test runs when the line plays, not when it was queued.
One clock	Recording time from packet arrival. Race time is a label on top of it.
Words are data	Every sentence shape lives in an editable file. Code chooses; it does not write.
Instruments that can fail	Every requirement has a detector or an expected outcome that a wrong build trips. Tests are written before features.

Decisions settled, 16 September 2026

ID	DECISION	OUTCOME	WHERE
DEC-1	File structure	One tool file for Mike to run. Words, config, roster and expected outcomes in data files beside it; the harness in tests/.	9
DEC-2	Audio and keys	No API keys in V3; manual audio with the timed-track kit. Baby Hoover runs to about V4, then Toddler Hoover brings in language models, with keys or local. V3 builds the seam for that now.	7.1, 7.4
DEC-3	Whose story	The story is always the human drivers. The camera may leave them for the lead or a critical event, but returns. The lead must be protected and can never be ignored.	6.1, 6.2
DEC-4	Restricted cars	Generic, not a person: any car can be restricted in any race. Naming falls back through roster match, then a descriptor, never "Player". Sholly was that race only.	7.3
DEC-5	Warnings	Silent for AI pairs; aired when a human is involved. Because the game reports no blame, "AI against a human" is judged by the penalty outcome.	5.3
DEC-6	Speed wording	Rounded. Default "three hundred and fifteen kilometres an hour", held as a setting in the words file so the broadcast-quality ruler can switch it in one edit.	7.3
DEC-7	Validation captures	A league race recorded with the current recorder (clean lights out, full finish) and a live spectator session with camera control on. Scheduled with Mike ahead of Pass 3.	9, 10

3 Architecture

Seven layers, each reading only the layer below. The race model is new. Everything above it is rebuilt to read it. Everything below it is V2's, with the source declared.



What happens to each V2 section

V2 SECTION	V3 FATE	WHY
1 Packet specification · 3 Capture writer · 9 Session folders	KEEP	Balanced integrity in every run. Writer loses markers (D-22); folders gain the source tag.
2 Synthetic input	EXTEND	Scancode sender kept; safety gate and circuit breaker wrapped round it.
4 State · 5 Parsing	REBUILD	Become the race model. Parser gains red flag, session start, final classification, conditions.
5b–5d Config, naming, roster, preflight	EXTEND	Roster becomes mandatory; broadcast name map loaded; one spoken form per race.
6 Pit Wall	REBUILD	Stakes and drama split; participation applies to drama only.
7 Suppression and fusion	REBUILD	Moves into the race model as contested positions and collapses from position data.
8 Slot grammar · 8b Scheduler, governor	EXTEND	Scheduler and burn ledger kept; words move to a file; claims checked at air time; filler engine replaces the governor's picker.
8c Gallery · 8d Booth	EXTEND	Hold floors and cut logging kept; director reads state and stakes; Focus Contract added.
11 Replay adapter · 12 Main	EXTEND	Source declared; timers on the packet clock.
13 Detectors	REBUILD	Move into the harness and check against the wire.
10 Simulator	RETIRE	The emulator and real captures replace it; simulated output is indistinguishable from capture (capture discipline).

4 Tier 1 — Know where the race is

4.1 The clock (R-01 to R-05, D-38 to D-40)

QUANTITY	DEFINITION
Recording time	Packet arrival time minus the recording-start instant. The only clock. Every artefact row carries it, plus unix time.
Race time	Recording time minus the start anchor. Blank before the anchor exists. A label, never used for scheduling.
Start anchor	The first lights-out event after a start-lights sequence in a race session.
Fallback anchor	Used only if no start-light or lights-out event arrives. It triggers when lap data shows the race running: every car's current lap has advanced past its grid lap and cars are moving at racing speed, for a sustained 3 s. Recorded as fallback. Never spoken as "lights out"; the Booth says "we're under way" once.
Restart markers	Any later lights-out in the same session. Listed, labelled restart, never the anchor.
Timers	All scheduling, cooldowns and decay run on packet arrival time, so fast replay behaves like recorded pace. Only the idle watchdog uses wall-clock time.

The Austria capture has no lights-out event, so the fallback is not hypothetical. Its test: replay Silverstone bin1 and Baku with the lights-out events hidden; the fallback must land within 1.5 s of the real event.

4.2 The race state machine (R-06, R-07, R-11)

STATE	ENTERED ON	LEFT ON
Pre-start	Race session opens	Formation lap flag set, or start lights begin
Formation	Session safety car status 3	Start lights begin. A safety car event with formation type returning or resuming is part of this state, never a restart call.
Start sequence	Start lights 1	Lights-out event → Green
Green	Lights out, or safety car resume	Safety car, virtual safety car, red flag, final lap, finish
Safety car · Virtual safety car	Safety car event deployed, type 1 or 2 (confirmed by session status)	Event returning, then returned or resume → Green, with a truthful call for each type
Red flag	Red flag event	Session-ended event while unclassified → Suspended
Suspended	As above. Idle watchdog extends to 10 min.	Session-started event → Restart grid
Restart grid	Session-started event after a suspension	Start lights → Start sequence, with the next lights out marked restart
Final lap	Leader begins the last racing lap. Racing laps count from the lap at the anchor, since the formation lap shifts numbering.	Leader's result status becomes finished → Finishing
Finishing	First car finished (lap data result status 3)	All running cars finished or retired, or final classification → Classified
Classified	Final classification decoded	Session ends → Closed

Every transition is logged with the packets that caused it. Race-winner and chequered-flag events are recorded as corroboration only, since they are player-relative.

4 Tier 1 — Know where the race is (continued)

4.3 What each state allows

CONTENT	PRE-START / FORMATION	GREEN	SAFETY CAR / VIRTUAL	RED FLAG / SUSPENDED	FINISHING / CLASSIFIED
Overtakes, battles, chase lines	—	Yes	Held ¹	—	Final crossings only
Lull material (filler)	Grid, humans' grid slots	Yes	Gaps, restart order	Order at stoppage, retirements	—
Results and winner	—	—	—	—	Yes
Camera	Grid and front	Director	Leader and restarters	Hold leader or incident	Leader, then humans crossing

¹ Position changes under a safety car are held and re-checked at restart; most are pit or order effects.

4.4 Car lifecycle (R-08, R-09, D-44, D-59)

Each car carries one state: **running**, **pit entry**, **in pit**, **pit exit**, **retired**, **disqualified** or **finished**. A resolver merges every signal for the same car inside a 5 s window: penalty type 16, the retirement event, lap-data result status 4, 5 or 7, and pit status. It produces **one** lifecycle change.

- **Retired is latched.** From then on the car can appear only in a retirement line, the stoppage order or the results. Any overtake, penalty, pit or battle claim naming it is dropped at source and logged.
- **Cause comes only from the data.** Final classification carries a result reason (terminal damage, mechanical failure, red flagged and so on). A cause is spoken only when that reason, or a detected contact on the car within 30 s, exists.
- **Pit shuffles are not overtakes or lead changes.** A position change where either car is in any pit state becomes a pit reorder, not a pass.

4.5 Finish and results (R-10, D-47, D-58)

1. **Winner:** the first car whose lap-data result status becomes finished, in a race session in Final lap. One line: the chequered flag and the winner, together.
2. **Humans' results:** each human's position as their status becomes finished, queued behind the winner call, one short line each.
3. **Classification:** the final classification packet is decoded (position, laps, grid, pit stops, result status and reason, total time, penalties). It confirms or corrects the results. A correction airs as a correction, never silently.
4. **After the flag:** no battle, chase or filler lines. Only results, and a closing line once classified.

4.6 Safety car taxonomy and pit cycles (R-11, R-12, D-61, D-68)

SAFETY CAR EVENT (TYPE · EVENT)	V3 CALL
Full · deployed	"Safety car." Enter Safety car state.
Virtual · deployed	"Virtual safety car." Never voiced as full.
Either · returning, then returned or resume	One call: "Safety car in this lap" at returning; "Racing again" at resume.
Formation · any	No safety car call. Part of Formation state.

Pit activity: pit entries and exits inside a 30 s window fuse into one strategic line ("Five cars in, Leclerc among them"). Exits are silent unless the car rejoins in a battle or ahead of a rival. A pit window with a human involved gets one line naming the human.

5 Tier 2 — Say only what is true

5.1 Derived passes and contested positions (R-13, R-14, D-33, D-41)

Passes come from the race position in lap data, sampled on every packet.

1. **Candidate:** car A moves ahead of car B, both running (not in any pit state, not retired), in Green.
2. **Confirmed pass:** the order holds for 2.0 s. The overtake event, if present, only sharpens the timestamp.
3. **Contested position:** two or more reversals within 20 s. Individual passes are withheld and one contested record is opened.
4. **Settled:** no reversal for 6 s. One line reports the outcome and the count ("They've swapped three times — Gasly keeps it").
5. **Collapse:** one car loses three or more places in 20 s without a pit state. It becomes one line naming *that* car, the places lost, and a cause only if detected (section 5.3). The individual passes are absorbed.
6. **Lead change:** a confirmed pass for P1 is a single lead-change call. There is no separate overtake line.

Starting values (2.0 s hold, 20 s contest window, 6 s settle, 3 places in 20 s) are provisional and tuned in Pass 1 against the Silverstone bin1 and Baku event streams, where the right answers are already known.

5.2 Claims checked at air time (R-15, R-17, D-42, D-48)

Every candidate line becomes a **claim**: what is asserted, about whom, the facts with the packets they came from, a validity test, and a maximum age. When the scheduler dequeues a claim, the validity test runs against the race model *now*.

CLAIM	STILL TRUE IF	IF NOT
Pass A on B	A still ahead of B, neither retired	Drop; a contested record may form
New leader X	X still leads	Drop
Quickest through the speed trap	Speed still equals the session best	Rewrite as "X through at ..." or drop
Battle A and B	Gap still under the battle threshold	Drop
Penalty for X	Always; age limit only	Past tense after 8 s (V2 behaviour kept)
Anything naming a car	Car not retired, unless the claim is its retirement	Drop

Drops and rewrites are logged with the reason. The harness counts them: a high drop rate means the queue is too long, not that the check is wrong.

5 Tier 2 — Say only what is true (continued)

5.3 Penalties and causes (R-16, R-18, D-43, D-49)

PENALTY TYPE	V3 TREATMENT
Warning	DEC-5. AI and AI: silent. Human and AI, both warned: one neutral line ("Contact between Kannedy and Hamilton, both warned"). The game reports no blame, so a warning never implies who caused it.
Penalty after contact	AI penalised, human not: "Penalty for Hamilton after that contact with Kannedy." A human penalised: always aired.
Time penalty	"Five-second penalty for X", with the seconds from the event, for the reason given where it is known.
Drive-through, stop-go	Named as such. Priority raised when a human or the lead is involved.
Grid penalty, disqualified	Named as such.
Retired (type 16)	Never a penalty line. Routed to the lifecycle resolver.
Lap invalidated, reminders, timers	Silent; logged.

Pass 1 builds this table from the appendix and verifies each type against the wire in the corpus (types 1, 4, 5 and 16 are already seen). "And it will stand" is removed: the data has no review state.

Cause rule. A cause slot accepts only a detected cause with provenance: a contact event involving the car, an off-track or corner-cutting penalty on it, or a final-classification result reason. Word banks never fill a cause. When no cause is detected, the line reports the fact alone ("Kannedy drops from third to seventh") without "the cause is not clear from here".

5.4 Detection that can fail (R-32, D-57)

The coverage check moves out of the tool and into the harness. It compares what aired against what the game sent, occurrence by occurrence. A red flag in the wire with no red flag line inside 10 s fails. So does a lead change in lap data with no lead-change line, or a winner with no winner line. Because it reads the wire, a parser that silently drops an event cannot hide from it.

6 Tier 3 — Point at what matters, safely

6.1 Humans are the story; the lead is protected (R-25, D-63, DEC-3)

The camera works in three layers, in this order of authority:

LAYER	WHAT IT DOES
1 • Protected moments	Always win: the start, every lead change, the final lap, the finish, and critical interrupts (major crash, safety car, red flag).
2 • Leader check-in	If the leader has not been on screen for 3 minutes of green-flag racing, the next shot change is a short shot of the leader.
3 • Human default	Otherwise the director chooses among the humans: their battles, gaps, recoveries. An away shot on AI cars must return to a human within 20 s unless a protected moment is still live.

Inside each layer the score keeps V2's split: **stakes** (leader, lead battle, podium) are never multiplied by participation; **drama** is, so humans' battles lift and AI-only battles damp. The Booth ties away shots back to the humans when true: "Norris leads, eight seconds up the road from Kannedy."

6 Tier 3 — Point at what matters, safely (continued)

6.2 Coverage floors and holds (R-26, R-27, D-28, D-50)

RULE	SPECIFICATION (STARTING VALUES)
Start	Lights out until the first corner clears (about 15 s): the front of the field. Incident phase, 2.5 s hold floor.
Lead change	Cut to the new leader within 3 s unless a critical interrupt is live.
Leader check-in	Leader unseen for 3 minutes in Green → one shot of 4–10 s.
Human default	Humans on screen for 60–70% of green-flag racing while any human is running.
Return to humans	Any away shot outside a protected moment returns to a human within 20 s.
Final lap	Leader for the last sector, unless a human is in a closer podium battle.
Finish	Leader at the crossing, held 5 s, then each human as they cross.
Red flag	Hold the incident or the leader; no cycling of parked cars.
Maximum hold	30 s in Green, 45 s in lulls, unless a rule above requires the shot. V2's phase floors (2.5 / 4 / 7 s) kept, start reclassified as incident.

DEC-3 calls this a fine balance still being worked. The 60–70%, 3-minute and 20-second values are the tuning knobs; Pass 2 reports each corpus race against them and any conflicts between layers.

6.3 Actuation safety (R-28 to R-30, D-64, D-65)

GUARD	SPECIFICATION
Source gate	Key presses only when the declared source is live game. Replays are always advisory. A replay with camera control requested logs one line saying why it was refused.
Window gate	Before every press: the foreground window belongs to the F1 25 process. If not, skip the press and count "focus lost", not a miss.
Spectating gate	The latest session data shows spectating is on (the flag, never the car index; per finding D-15/D-16).
Circuit breaker	Three consecutive failed selections, or no success in the first 60 s: stop pressing, switch to advisory, log "camera unreachable" once, and retry the gates every 60 s.
Honest observation	External camera moves count as operator override only when source and actuation are both live. In a replay the recorded camera is data.
Honest reporting	Manifest actuation state is one of: live-controlled, advisory by choice, advisory because unreachable, advisory because replay. Hit and miss rates exist only for live-controlled.

6.4 The Focus Contract (R-31)

The decided contract, now built. Every line records **who is on screen, how long the shot is expected to stay, and whether the line's subject is that car**.

- **Voice:** on-screen subject goes to the lead; off-screen to the analyst.
- **Phrasing:** off-screen lines use a variant with the full name and a signpost ("Further back, ...").
- **Length:** an on-screen line must fit the shot's remaining expected hold, or a shorter variant is chosen.
- **Cut requests:** the Booth may ask for a cut; the Gallery may decline while mid-hold on a higher score. Both are logged.

The focus log is a new artefact, and test T-04 becomes runnable.

7 Tier 4 — Sound like a broadcast

7.1 The words file (R-19, R-21, D-45, D-60)

All sentence shapes move to `hoover_lines_v3.json`, beside the config and roster. Each claim kind lists variants. The code picks one that fits; it never edits a sentence.

```
"PASS": {
  "requires": ["attacker", "defender"],
  "optional": ["position"],
  "variants": [
    {"id": "pass.01", "text": "{attacker} gets by {defender}", "screen": "on"},
    {"id": "pass.02", "text": "{attacker} goes through on {defender} for {position}", "screen": "on"},
    {"id": "pass.03", "text": "Further back, {attacker_full} takes {defender}", "screen": "off"}
  ],
  "past": {"gets by": "got by", "goes through": "went through", "takes": "took"},
  "max_age_s": 8
}
```

RULE	BEHAVIOUR
Fit, never chop	The chooser takes variants that fit the word budget, the on/off-screen frame and the remaining hold. If none fits, the claim is dropped and logged. Truncation code is deleted.
Every kind reachable	The harness fails if a defined kind or variant never aired across the corpus without a logged reason (for example, no pit windows in a 5-lap race).
Rotation, not prefixes	The V2 burn ledger now keys on variant ID and subject. "Still", "Again", "Back to" prefixes are removed.
Voice idiom separate	Lead and analyst have their own variant sets; the analyst never gets play-by-play shapes.
Toddler Hoover seam	Claims already carry verified facts, their provenance and a validity test: the right input for a language model. In Toddler Hoover a model may rephrase variants marked <code>rewrite_ok</code> from those facts only. It never decides facts, and the template is the fallback if it is slow or absent.

7.2 Filler from real material (R-20, D-45, D-46)

V2 had seven filler kinds and could only reach one. V3 replaces the picker with a **lull engine**: when the channel has been quiet past the silence threshold, it asks the race model what is true and interesting now, and picks the best available material.

MATERIAL	AVAILABLE WHEN	EXAMPLE LINE
Gap trend	A gap under 2 s changing by 0.2 s a lap or more	"Tsunoda has taken half a second out of Kannedy in two laps."
Human check-in	A human not mentioned for 90 s	"Kannedy running sixth, three seconds off Norris."
Laps remaining	Final third of the race	"Four laps to go."
Recovery thread	A car that collapsed earlier and has regained places	"Lababera back up to eleventh after that early drop."
Pit window	Pit stops underway or due	"Leclerc is the first of the leaders to stop."
Conditions	A verified change in weather or temperature (section 8)	"Light rain now reported."
Stoppage order	Suspended state only	"Under the red flag, Piastrì leads Leclerc and Gasly."

If no material qualifies, the Booth stays silent. **Silence is allowed; repetition is not.** The lull engine is gated by race state (section 4.3) and never fires after the flag.

7 Tier 4 — Sound like a broadcast (continued)

7.3 Pacing, names and numbers (R-22 to R-24, D-40, D-51, D-66, D-67)

AREA	SPECIFICATION (STARTING VALUES)
Occupancy ceiling	Estimated speech no more than 75% of any 60 s window. Past it, only interrupt-tier claims are admitted.
Breathing room	After 4 back-to-back lines or 12 s of continuous speech, a 1.0 s gap is forced. Austria ran at 80% back-to-back; V3's target is under 50%.
Roster required	No roster file means a loud stop at start with the path expected. It never silently degrades, as it did in three V2 races.
Broadcast names	The decided map is loaded (Kannedy, Kannedy Jr, Meadows, Osterman, Lababera). One spoken form per driver per race, fixed at first mention, capitalised at sentence start.
Restricted cars	DEC-4: generic. Restricted telemetry still sends race number, team and human flag. Ladder: (1) roster match on race number <i>and</i> team, plus network ID where present, gives the broadcast name; any ambiguity refuses the match. In Austria, Faze and Raider both ran number 2. (2) Otherwise a descriptor, "the Red Bull, number 76". Never "Player", never a car index. A restricted human stays human in the participation model.
Speech normaliser	Runs last, before output. Speeds rounded and written out, default "three hundred and fifteen kilometres an hour" as a words-file setting (DEC-6); penalty seconds and laps remaining written out; positions as "sixth" in prose, "P6" never spoken; no abbreviations.
Pronunciation	The lexicon keeps only entries that change sound. Name-to-capitalisation aliases are removed.

7.4 The timed-track kit — manual audio (R-05, parked brief folded in)

Per DEC-2, V3 stays keyless but produces everything needed to turn the script into one aligned audio track without placing lines by hand.

1. **Subtitle file.** Every line at recording time, plus >> LIGHTS OUT and >> RESTART LIGHTS OUT cues. Dropped on CapCut at 00:00, it shows alignment before any audio exists.
2. **Blocks.** Lines less than 2 s apart grouped into blocks, each a clean speaker-tagged text file named by block and time, ready for Text-to-Dialogue. About 36 for a 5-lap race rather than 94 lines.
3. **Mix mode.** `--mix <session> --clips <folder>` places each exported WAV at its recording time on one track starting at 00:00. Clips that run long push the next one back, with the lateness logged.
4. **Calibration.** The mix report fits real seconds per word for each voice and writes a calibration file. The next run schedules with real durations. The 0.344 s estimate retires after one trial (closes test T-03).
5. **Synthesis interface.** Estimate, manual kit, batch and live all implement one interface. Batch and live are stubs until DEC-2 changes, but plugging one in changes no scheduling code.

In CapCut the whole job is: drop WAV and subtitle file at 00:00, check the lights-out cue meets the visible lights, slide once if OBS started late.

8 Tier 5 — Instruments that can fail

ITEM	SPECIFICATION
Declared source (R-33)	--source live replay fast, required. Written to run summary, manifest, script header and every artefact. V2's mode: live field is removed.
Replay parity (R-34)	Fast and recorded-pace replay of the same capture must produce the same claims, in the same order, within 0.25 s, finish included. A harness check, run on every pass.
Artefact schema v3 (R-35)	Cuts: t_rec, t_race, held_s as seconds (blank if open), source, actuation state. Script header states the anchor source truthfully. Log names delimited from scores.
Capture hygiene (R-36)	Per-type counts reset at session open. Markers go to a sidecar index file, never into the capture.
Conditions (R-37)	Session weather, track and air temperature, and the rain forecast samples decoded and verified by wire readback on s02 and s04 (closes T-01). Available to Pit Wall and the lull engine; state changes only, no invented narrative.
Session guard (R-38)	A session opens only on sustained lap data from a full grid, not on a post-race trickle.
Format isolation	All F1 25 packet layouts in one swappable table, per the AIRAC decision. V3 adds no format knowledge outside it.

Pass 0 must prove the harness can fail

Before any V3 feature is built, V2 is run through the harness on the five corpus races. The expected result is a specific set of failures. Examples:

RACE	V2 MUST FAIL
Silverstone bin1	A15 (start call at 2.1 s), A18 (Colapinto "passes" Gasly while behind), A6 (winner)
Baku	A6 (red flag), A16 (11 lines while suspended), A22 (three retirements as penalties)
Austria	A6 (winner never named), A21 (Gasly on screen at the flag), A16 (filler after the flag)
Silverstone s04	A17 (Bearman lines after retirement), A19 ("closing the door."), A20 (key presses on a replay)
Silverstone s02	A21 (Valor on screen as Norris wins), A19 (pit-lane line cut off, to be confirmed by T-09)

If V2 passes any of these, the detector is wrong and is fixed before Pass 1. **A harness V2 can pass is not a harness.**

8 Tier 5 — The harness and its detectors

The harness is a separate test file. It replays a capture through V3, then checks the outputs against two things: the **wire** (what the game actually sent) and an **expected-outcomes file** per race, written from the V2 analysis where the right answers are already established.

Detectors

A29–A31 (red) were added in V1.2. They were found while writing the Pass 0 expectations: the analysis showed those failures and no listed detector caught them.

ID	FAILS WHEN	GUARDS
A1	Two lines overlap in estimated speech (kept from V2)	Scheduler
A6	A coverage-floor event in the wire (red flag, lead change, winner, safety car) has no matching line within 10 s — per occurrence	R-32
A7	A raw gamertag reaches a line (kept)	R-23
A11	Two mutually exclusive states aired within 1 s (kept)	R-15
A14	A cause asserted without detected provenance (kept, widened)	R-18
A15	A start call airs before the lights-out event, or more than one per start	R-04
A16	A battle, chase or pass line airs in Suspended, Red flag, Finishing or Classified	R-07
A17	Any line other than retirement or results names a retired car after its retirement	R-08
A18	A claim is false at its air time against lap data (pass order, leader, quickest)	R-15
A19	A line ends in a dangling word or fails the no-truncation check	R-21
A20	A key press is attempted with source not live, window not F1 25, or spectating off	R-28
A21	The leader is not on screen at the finish crossing	R-26
A22	A penalty line for type 16, or "penalty" wording for a warning	R-16
A23	Occupancy over 75% in a 60 s window, or a back-to-back chain past the limit	R-22
A24	Fast and recorded-pace outputs diverge by more than 0.25 s or in content	R-34
A25	A retirement is reported by more than one line	R-09
A26	An away shot lasts past 20 s outside a protected moment, or human share leaves its band	R-26
A27	"Player", a car index, or an ambiguous roster match reaches air	R-23
A28	The leader goes unseen for more than 3 minutes of green-flag racing	R-26
A29	The last line about a pair that swapped late disagrees with their classified order	R-14
A30	A formation or resume transition voiced as a safety car restart, or a virtual safety car voiced as full	R-11
A31	More than 4 pit lines in any 60 s	R-12

9 Build plan

Four passes, one Claude Code brief each. Within a pass Claude Code builds in **stages** and gates each stage on the synthetic races before starting the next, so nothing is built blind. Mike runs the real-capture gate **once per pass**; Dustin judges where taste matters.

PASS	BUILDS, BY INTERNAL STAGE	REQUIREMENTS	CHECKPOINT (MIKE'S RUN)
0 • Harness	Test file, independent decoder, wire truth model, detectors A1–A31 and H1–H3, expected outcomes for five races, synthetic fixture corpus. <i>Built and verified 16 September.</i>	R-32 (partial)	V2 fails exactly where the analysis says: GATE: PASS
1 • Race model	1a Foundation: declared source, one clock, start anchor and fallback, race state machine, car lifecycle, finish and final classification, red flag, packet-clock timers, artefact schema v3, subtitle file, session guard. 1b Truth: derived passes, contested positions, collapses, lead-change merge, claims checked at air time, penalty classifier, cause rule.	R-01–R-04, R-06–R-18, R-33–R-35, R-38	A6, A11, A14–A18, A22, A24, A25, A29, A30 green on all five races. bin1: Gasly ahead, Ronin P6. Baku: three retirements, red flag, restart. Fallback anchor within 1.5 s.
2 • Broadcast	2a Camera: protected moments, leader check-in, human default and return rule, stakes and drama split, holds, actuation gates, circuit breaker, Focus Contract log. 2b Voice: words file and chooser, no truncation, lull engine, pacing, roster and restricted-car naming, speech normaliser, blocks, mix mode, calibration. 2c Hygiene: counts reset, markers to sidecar, conditions decode, format isolation, simulator retired.	R-05, R-19–R-31, R-36, R-37	A7, A19–A21, A23, A26–A28, A31, H2, H3 green. Human share in band. Zero key presses on replay. Weather read back. Dustin judges the camera balance and scripts, and makes the first manual audio track.
3 • Validation	Mike: full corpus through the emulator; one live spectator session with camera control on and OBS running; a league race recorded with a clean lights-out event; files for the final audio trial.	All; T-03, T-11	Harness green on every capture including the new ones; live direct select 97.8% or better; one aligned audio track with one slide at most.

Why four passes, not one and not seven

CHOICE	REASON
Not one build	Layers depend on each other: the camera needs lead changes and the finish; the words need the camera's shot record. One giant build that fails on real data gives no way to tell which layer is wrong. V2 was built in one go.
Not seven	Mike's run is the bottleneck, and most internal steps don't need it. A real-capture checkpoint belongs only where real data can disagree with the synthetic races, or where Dustin must judge.
Harness first	V2 reported health while wrong. Every later checkpoint depends on detectors that have already caught V2.
Race model as one pass	States and truth answer one question: what is true about the race. Neither can be proven on real races without the other.
Broadcast as one pass	Camera and voice are coupled by the Focus Contract; hygiene changes no broadcast behaviour and rides along safely.

Each brief follows the Pass 0 brief's shape: scope, specification by stage, synthetic-race tests Claude Code must pass before pushing, real-capture tests for Mike, out-of-scope list, hand-back, and a one-page Mike sheet with run 1 and, if needed, run 2.

10 Expected outcomes and acceptance

These are the headline checks per race, taken from the V2 analysis. The full expected-outcomes files are written in Pass 0.

RACE	V3 MUST
Silverstone bin1	One start call at lights out (± 0.3 s). Colapinto–Gasly reported as contested, with Gasly ahead. Ronin's final-lap pass on Tsunoda aired, Ronin P6. VaLoR's lap-1 retirement as one retirement line, no pit line. Winner and chequered flag in one call within 3 s of the leader finishing. Zero lines after the results.
Baku	Virtual safety car called as virtual. Red flag called within 3 s. Zero track-action or filler lines while suspended; stoppage order allowed. Restart grid, then a restart lights-out call. Pure, Valor and Ronin each get exactly one retirement line and zero penalty lines. Stop-go and 10-second penalties named correctly. Session ending under the virtual safety car handled without idle-close errors.
Austria	Fallback anchor used and declared. No "safety car in" call at the formation end. Lead changes between Faze and Valor as single calls, no doubling. Valor named winner; Osterman, Kannedy Jr and the rest of the humans' results read. Leader on screen at the flag. Restricted car 76 named by roster match on number and team, or "the Red Bull, number 76"; never "Player". Faze and Raider, both number 2, never confused. Car 76's collapse as one line.
Silverstone s04	One start call at the real start; any later lights out marked restart. Bearman: one retirement line, zero lines about him afterwards. Pit cycle: no more than 4 pit lines in any 60 s, no "new leader" from pit order. No line cut off. Winner and flag together. Zero key presses (replay).
Silverstone s02	Norris on screen at the finish. Leader seen at least every 3 minutes. Valor on screen 60–70% of green-flag racing while running, and every away shot returns to him within 20 s. No pit line cut off.

Overall acceptance for V3

MEASURE	TARGET
Harness on the five-race corpus	All detectors green, fast and recorded pace
False statements (A17, A18, A22)	Zero
Coverage events without a line (A6)	Zero
Busiest 60 s occupancy	75% or less, Austria included
Largest share for any single variant	15% of lines or less
Leader on screen at each finish	Every race with a flag
Human share, green-flag racing	60–70% while humans are running
Key presses on replay	Zero
Live direct select (Pass 3)	97.8% or better
Manual audio trial (Pass 2 first, Pass 3 final)	One track, one slide at most, no per-line placement

Not an acceptance measure yet: "sounds good". Quality against the professional-broadcast ruler is the bar, but the ruler is still being built. V3 acceptance is truth, coverage and mechanics. The ruler scores V3's output once it exists, and that score shapes V4.

11 Risks

RISK	EFFECT	MITIGATION
Derived passes miss real ones	A 2 s hold drops brief passes and re-passes on the same straight	Tune on bin1 and Baku, where the events are known; keep the overtake event as timing evidence
State machine meets an unseen sequence	A new combination (black flag, disqualification, abandoned race) lands in no state	Unknown transitions go to a logged holding state that allows only safe content; the harness fails on any holding-state entry
Replay is not live	Actuation, focus and timing under OBS load are untested until Pass 3	Pass 3 live session is a hard gate; Step 0.5 load test runs in the same sitting
File size	One file of 6,000 lines or more becomes hard to brief and review	DEC-1: data files carry the bulk; revisit a /lib/ split only if a pass brief cannot fit
Bigger passes, bigger fix loops	A failed Mike run on Pass 1 or 2 covers more code	Stage gates on synthetic races before push; each detector points at one requirement, so a failure names its layer
Human default meets the lead	A race where the humans are far from a busy lead fight strains the 60–70% target	Protected moments first, check-ins next, human default last; conflicts logged; Pass 2 reports each race
Expected outcomes are wrong	The harness enforces a mistake from the analysis	Each expected outcome cites its evidence; any V3 disagreement is investigated in the wire before the expectation changes
Mike's calendar	Pass 3 slips	Passes 0–2 need no live game, only replays; DEC-7 captures scheduled ahead of Pass 3
Game patch moves offsets	Decoding breaks mid-build	Format table isolation (section 8); the harness catches decode drift on the first run

12 Next steps

1. **Dustin and Mike:** run Pass 0 on the real captures; upload `pass0_reports_run1.zip`.
2. **Claude:** review the reports with Dustin. On GATE: PASS, write the Pass 1 brief (race model) and its Mike sheet.
3. **Claude Code:** build Pass 1 in stages 1a and 1b, each gated on the synthetic races; push.
4. **Claude, then Mike:** verify the push, then Mike's Pass 1 run.

A Requirement traceability

REQ	REQUIREMENT	SECTION	PASS	PROVEN BY
R-01	One clock from recording start	4.1	1	A24, bin1 expected
R-02	Start anchor is first real lights out	4.1	1	A15, Baku expected
R-03	Declared silent fallback	4.1	1	Hidden-event test
R-04	One start call per start	4.1	1	A15
R-05	Renderable to a timed track	7.4	2 - 3	Audio trial, T-03
R-06	Race state model	4.2	1	Transition logs, A16
R-07	Content gated by state	4.3	1	A16
R-08	Latched car lifecycle	4.4	1	A17
R-09	One retirement, one line	4.4	1	A25, A22
R-10	Finish, winner, results	4.5	1	A6, Austria expected
R-11	Truthful safety car, virtual safety car, red flag, restart	4.2, 4.6	1	A6, A30
R-12	Pit activity fused	4.4, 4.6	1	A31, s04 expected
R-13	Derived passes	5.1	1	A18
R-14	Settled outcomes	5.1	1	A29, bin1 expected
R-15	Air-time validation	5.2	1	A18, A11
R-16	Penalty meaning	5.3	1	A22
R-17	One event, one line	5.1, 5.2	1	Austria expected
R-18	No undetected causes	5.3	1	A14
R-19	Words in data, all reachable	7.1	2	Reachability check
R-20	Filler from real material	7.2	2	Variant share $\leq$ 15%
R-21	Nothing cut off	7.1	2	A19
R-22	Pacing ceilings	7.3	2	A23
R-23	Roster names, one form, restricted ladder	7.3	2	A7, A27
R-24	Speech-ready numbers	7.3	2	Normaliser tests
R-25	Stakes never multiplied	6.1	2	s02 expected
R-26	Protected moments, human default	6.1, 6.2	2	A21, A26, A28
R-27	Holds and start phase	6.2	2	Cuts log
R-28	Gated key presses	6.3	2	A20
R-29	Circuit breaker	6.3	2 - 3	Live session
R-30	Honest observation and rates	6.3	2	Manifest check
R-31	Focus Contract logged	6.4	2	T-04
R-32	Wire-based checks	5.4, 8	0	Pass 0 gate
R-33	Declared source	8	1	Manifest check
R-34	Faithful fast replay	8	1	A24
R-35	Honest artefacts	8	1	Schema check
R-36	Counts reset, no markers	8	2	Integrity check
R-37	Conditions decoded	8	2	T-01 readback
R-38	No trickle sessions	8	1	bin1 expected

B Defect transfer to V3

V2's open defects transfer to V3 explicitly, per the supersession rule. Each is assigned to the pass that closes it, or retired with a reason.

PASS	DEFECTS TRANSFERRED
Pass 0 · Harness	D-57 coverage checker blind (its V3 replacement lives in the harness)
Pass 1 · Race model	D-38 guessed anchor · D-39 early start call · D-44 retirements as pit or penalty · D-46 talking over stoppages and after the flag · D-47 no ending · D-52 artefact columns · D-53 fast replay parity · D-54 spurious session · D-56 red flag unhandled · D-58 final classification discarded · D-59 no retired latch · D-61 pit ticker · D-65 source not declared · D-68 safety car taxonomy · D-25 log delimiter · D-37 no result readback · D-32 arbiter completion · D-33 evented overtakes · D-34 one beat per participant · D-36 penalty generators · D-41 suppression drops outcome · D-42 no air-time check · D-43 penalty classes flattened · D-48 lead change doubled · D-49 collapse misattributed, cause bank
Pass 2 · Broadcast	D-27 opening shot · D-28 finish coverage · D-50 over-holding · D-63 participation scales stakes · D-64 actuation safety · D-40 not renderable, saturation · D-45 filler loop · D-51 broadcast names · D-55 lexicon aliases · D-60 truncation · D-66 roster not loaded · D-67 numbers and units · D-20 run-cumulative counts · D-22 markers in capture · D-62 weather unused
Pass 3 · Validation	D-23 packets between sessions · D-24 failed start orphan folder · D-26 camera actuation · D-29 left-the-track cuts · D-31 restricted car direction (all not testable on replay)
Retired	D-30 two-second shots (fixed in V2, guarded by the harness) · D-35 spurious safety car (explained; superseded by D-68)

Open tests from the V2 analysis

TEST	CLOSED IN
T-01	Pass 2, conditions readback
T-02	Pass 0 truth report, confirmed by Pass 1 state log on s04
T-03	Pass 2 first audio trial and calibration; Pass 3 final
T-04	Pass 2, Focus Contract log
T-05	Pass 1, s04 replayed through the emulator in advisory mode
T-06	Pass 2, VI's spectated camera from the Austria capture against V3's director
T-07, T-10	Pass 0, observed in the harness reports
T-08	Pass 1, cause provenance logged
T-09	Pass 0, observed in the harness reports
T-11	Pass 3, live spectator session